AI-Based Fuel Consumption Prediction Using Linear Regression

Detailed Project Plan and Product Backlog

Project Title

AI-Based Fuel Consumption Prediction Using Machine Learning (Linear Regression)

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1. Project Overview

Project Description

Fuel consumption prediction is one of the important applications of Artificial Intelligence in the automobile industry. The objective of this project is to build an AI model that predicts the fuel consumption of vehicles based on different vehicle characteristics such as engine size, cylinders, horsepower, vehicle weight, transmission type, and model year.

The project uses **Linear Regression**, a supervised machine learning algorithm, to estimate the amount of fuel consumed by a vehicle. The trained model can help manufacturers, vehicle owners, and environmental agencies improve fuel efficiency and reduce carbon emissions.

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2. Project Objectives

- Predict fuel consumption accurately.
- Reduce manual estimation errors.
- Analyze vehicle parameters affecting fuel consumption.
- Build a machine learning model.
- Deploy the model using Streamlit.
- Visualize data using graphs.

- Evaluate model performance.

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3. Problem Statement

Fuel consumption depends on many factors including engine size, cylinders, horsepower, vehicle weight, and driving conditions. Manual estimation is difficult and inaccurate.

The aim is to build an AI-powered prediction system that estimates fuel consumption using historical vehicle data.

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4. Scope

Included

- Data Collection
- Data Cleaning
- Data Analysis
- Feature Engineering
- Model Training
- Prediction
- Evaluation
- Deployment

Not Included

- Real-time IoT data
- GPS Tracking
- Deep Learning
- Mobile Application

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5. Technologies Used

Category	Tool
Programming	Python
IDE	Google Colab
ML Library	Scikit-learn
Data Analysis	Pandas
Numerical	NumPy
Visualization	Matplotlib
Charts	Seaborn
Model	Linear Regression
Deployment	Streamlit
Version Control	GitHub

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6. Team Members

Role	Responsibility
Project Manager	Planning
Data Analyst	Data preprocessing
ML Engineer	Model building
UI Developer	Streamlit Application
Tester	Testing
Documentation	Report Preparation

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7. Project Deliverables

- Dataset
- Python Source Code
- Trained Model
- Streamlit Web App
- Final Report
- GitHub Repository

- Presentation

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8. Work Breakdown Structure (WBS)

Phase 1

Project Initiation

- Requirement Gathering
- Problem Definition
- Goal Identification

Phase 2

Planning

- Dataset Selection
- Timeline
- Risk Analysis
- Resource Planning

Phase 3

Data Collection

- Download Dataset
- Verify Data
- Store CSV

Phase 4

Data Preprocessing

- Missing Value Handling
- Duplicate Removal
- Data Cleaning
- Feature Selection

Phase 5

Exploratory Data Analysis

- Correlation Matrix
- Histograms
- Scatter Plots
- Boxplots

Phase 6

Model Development

- Train Test Split
- Linear Regression
- Model Training
- Model Saving

Phase 7

Evaluation

- MAE
- RMSE
- R^2 Score

- Prediction Graph

Phase 8

Deployment

- Streamlit UI
- User Inputs
- Prediction Result

Phase 9

Testing

- Functional Testing
- Integration Testing
- Performance Testing

Phase 10

Documentation

- User Manual
- Final Report
- GitHub Upload

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9. Project Timeline (8 Weeks)

Week	Activity
Week 1	Requirement Analysis

Week	Activity
Week 2	Dataset Collection
Week 3	Data Cleaning
Week 4	Data Visualization
Week 5	Model Training
Week 6	Model Evaluation
Week 7	Streamlit Deployment
Week 8	Documentation & Presentation

PRODUCT BACKLOG (Detailed)

Epic 1 – Project Setup

Product Backlog ID	User Story	Priority	Story Points	Status
PB-01	As a developer, I want to create a GitHub repository so that I can manage the project code.	High	2	To Do
PB-02	As a developer, I want to create the project folder structure so files remain organized.	High	2	To Do
PB-03	As a developer, I want to install Python libraries so the project dependencies are ready.	High	3	To Do

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Epic 2 – Dataset Management

ID	User Story	Priority	Story Points
PB-04	Import dataset into Google Colab.	High	3
PB-05	Validate dataset columns.	High	2

ID	User Story	Priority	Story Points
PB-06	Handle missing values.	High	3
PB-07	Remove duplicate records.	Medium	2
PB-08	Save cleaned dataset.	Medium	2

Epic 3 – Data Analysis

ID	User Story	Priority	Story Points
PB-09	Generate summary statistics.	High	2
PB-10	Plot histograms for all numerical features.	Medium	3
PB-11	Create scatter plots between key features and fuel consumption.	High	3
PB-12	Generate a correlation heatmap.	High	2
PB-13	Detect outliers using boxplots.	Medium	3

Epic 4 – Machine Learning Model

ID	User Story	Priority	Story Points
PB-14	Split dataset into training and testing sets.	High	2
PB-15	Train a Linear Regression model.	High	5
PB-16	Generate predictions on the test set.	High	3
PB-17	Save the trained model using `joblib`.	High	2
PB-18	Load the trained model for inference.	Medium	2

Epic 5 – Model Evaluation

ID	User Story	Priority	Story Points
PB-19	Calculate MAE.	High	2
PB-20	Calculate MSE.	High	2
PB-21	Calculate RMSE.	High	2
PB-22	Calculate R^2 Score.	High	2
PB-23	Compare actual vs predicted values.	Medium	3

Epic 6 – AI Prediction Interface

ID	User Story	Priority	Story Points
PB-24	Build a Streamlit interface.	High	5
PB-25	Add input fields for vehicle features.	High	3
PB-26	Display predicted fuel consumption.	High	2
PB-27	Validate user input values.	Medium	3
PB-28	Improve UI layout and styling.	Low	2

Epic 7 – Visualization

ID	User Story	Priority	Story Points
PB-29	Plot actual vs predicted graph.	Medium	3
PB-30	Plot feature importance or coefficients.	Medium	2
PB-31	Display regression line visualization.	Medium	3
PB-32	Save graphs as image files.	Low	2

Epic 8 – Testing

ID	User Story	Priority	Story Points
PB-33	Perform unit testing for data preprocessing.	Medium	3
PB-34	Test prediction accuracy.	High	3
PB-35	Test Streamlit application functionality.	High	3
PB-36	Fix identified bugs.	High	5

Epic 9 – Documentation

ID	User Story	Priority	Story Points
PB-37	Write installation guide.	Medium	2
PB-38	Create project report.	High	5
PB-39	Prepare presentation slides.	Medium	3
PB-40	Update GitHub README.	High	2

Sprint Plan

Sprint	Duration	Backlog Items
Sprint 1	Week 1–2	PB-01 to PB-08 (Project Setup & Dataset)
Sprint 2	Week 3–4	PB-09 to PB-18 (EDA & Model Development)
Sprint 3	Week 5–6	PB-19 to PB-28 (Evaluation & Streamlit UI)
Sprint 4	Week 7–8	PB-29 to PB-40 (Visualization, Testing & Documentation)

Conclusion

The **AI-Based Fuel Consumption Prediction Using Linear Regression** project successfully demonstrates how Artificial Intelligence and Machine Learning can be used to estimate vehicle fuel consumption based on important vehicle characteristics such as engine size, number of cylinders, horsepower, vehicle weight, transmission type, and other related features. By using historical vehicle data and applying the Linear Regression algorithm, the system learns the relationship between these input parameters and fuel consumption, enabling accurate and efficient predictions.

Throughout the project, the complete Machine Learning lifecycle is followed, including data collection, data preprocessing, exploratory data analysis (EDA), feature selection, model training, model evaluation, and deployment using a Streamlit web application. Evaluation metrics such as **Mean Absolute Error (MAE)**, **Mean Squared Error (MSE)**, **Root Mean Squared Error (RMSE)**, and **R^2 Score** are used to assess the model's performance and ensure that the predictions are reliable